// VERSION 1.0.0

PANINI: UI/UX Design System

This design system outlines the visual language and interaction guidelines for the PANINI voice-first assistant. By marrying the conceptual depth of Pāṇini's ancient algorithmic grammar (Sutras) with the robust accessibility and structural clarity of Google's Material Design principles, we establish a paradigm termed **"Algorithmic Antiquity."**

The interface abandons the conventional "sci-fi neon blue" for a grounded, precise palette of Obsidian, Copper, and Gold. It is designed for real-time observability, ensuring users understand exactly how PANINI delegates tasks across its four core capabilities: Search, Browser, Desktop, and Code.

Core Design Principles (Google M3 × Pāṇini)

1. Material is the Metaphor (The Dhatu)

Instead of paper and ink, our "material" is nodes and threads (Sutras). Information spaces are constructed from structural roots (Dhatu) expanding into complex derivation trees. UI elements behave with physical predictability.

2. State Visibility (System Status)

Google's UX heuristic of "Visibility of System Status" is paramount. The Electron avatar and the web console must always clearly indicate whether PANINI is listening, parsing, routing, or executing via constant, non-intrusive visual feedback.

3. Motion as Meaning (Vriti)

Animations are not decorative; they are functional. When a spoken command is parsed, motion directs the user's eye from the central input node to the target capability (e.g., a thread animating toward the Browser quadrant).

4. Intentional Hierarchy (Aṣṭādhyāyī)

Layouts follow a strict hierarchical order. The Web Console uses a high-contrast structural grid. Secondary data (logs, memory) is visually demoted, while active execution (Code generation, Desktop observation) dominates the view.

THEME DEFINITION: ALGORITHMIC ANTIQUITY

The visual aesthetic relies heavily on high-contrast, dark mode execution. We utilize geometric sans-serifs for primary legibility, paired with strict monospace typefaces for telemetry and runtime output. Subtle Devanagari structural motifs can be used strictly as background watermarks or section identifiers to reinforce the brand identity.

The Avatar: Interaction States

The PANINI Avatar is an Electron overlay residing on the Windows 11 desktop. It is the controller and the primary point of voice interaction. It eschews anthropomorphic shapes in favor of an abstract, breathing geometric core.



IDLE (Root)



LISTENING



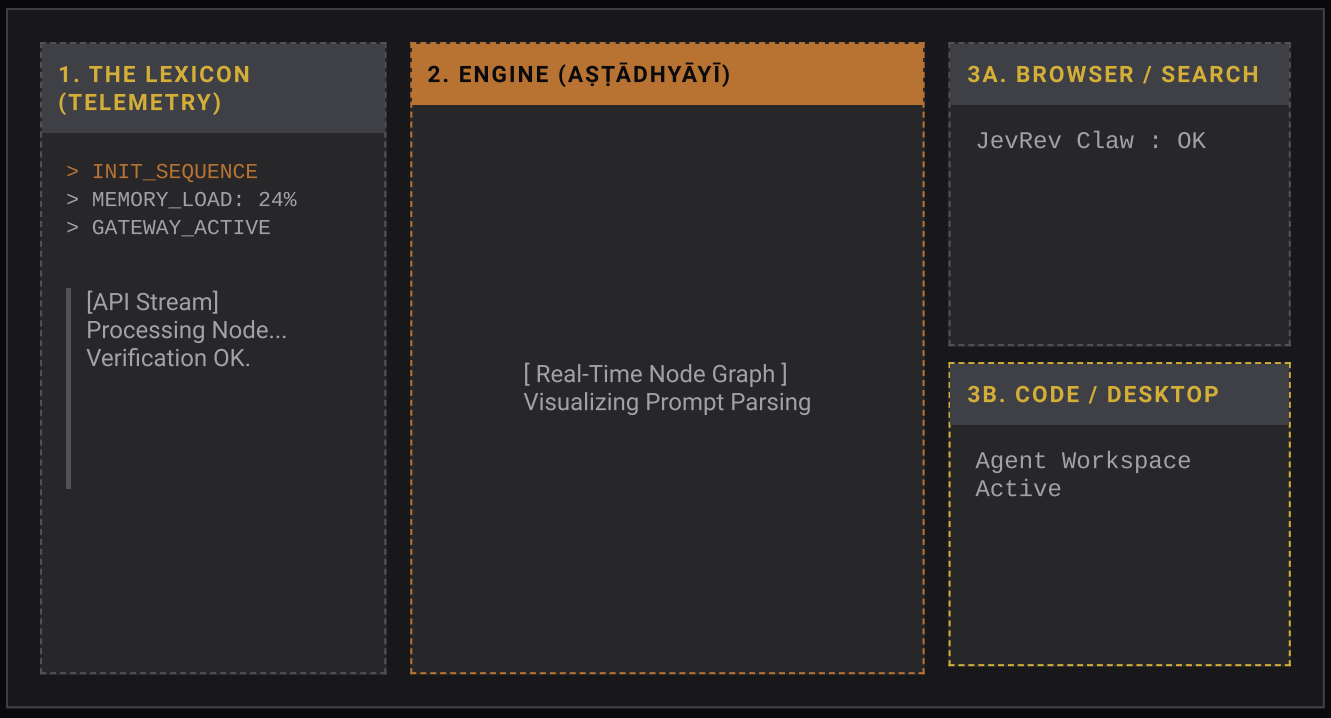
ROUTING (Sutra)

AVATAR STATE DEFINITIONS

STATE	VISUAL BEHAVIOR	GOOGLE UX PRINCIPLE APPLIED
Idle (Dhatu)	A slow-pulsing, minimal copper core. Opacity transitions subtly from 60% to 100% every 4 seconds.	**Non-intrusive presence.** Allows users to maintain focus on primary Windows applications.
Listening	Core expands. A mandala-like waveform reacts to vocal amplitude. Colors shift to bright Amber.	**Clear affordance & feedback.** The user instantly knows the mic is hot and registering input.
Parsing	Waveform snaps into concentric geometric rings rotating in opposite directions.	**System Status.** Indicates processing time gracefully without freezing the UI.
Routing (Delegation)	Core fires a directed "thread" to a designated outer node (representing Browser, Code, Search, Desktop).	**Spatial meaning.** Reinforces the system architecture through visual direction.

The Web Console (localhost:3200)

The web console acts as the "JARVIS-style" command center. While the desktop avatar handles immediate input, the web console provides deep observability into the Runtime :3200 gateway. It is designed as a dense, high-information dashboard.



CONSOLE COMPONENT BREAKDOWN

1. Left: Telemetry & Memory

Streams raw data, API calls, and system health. Uses strict monospace formatting to align with terminal aesthetics. Anchors the technical reality of the system.

2. Center: The Engine Visualizer

The focal point. Uses WebGL/Canvas to display the derivation tree of the user's intent. When PANINI solves a problem, this area animates the resolution structure.

3. Right: Capability Quadrants

Monitors the 4 active agents. If the Desktop agent is using the named pipe to a Windows app, its quadrant highlights with a copper border to indicate active focus.

Bilingual Touches (Micro-copy)

Use faint Devanagari subscripts (e.g., अन्वेषण for Search) under English titles to reinforce the PANINI lore without disrupting standard legibility.

Design Tokens & Specifications

To maintain consistency between the Node.js/React web console and the Electron runtime, strictly adhere to these CSS/theme definitions. The aesthetic balances high legibility with the warmth of ancient metallurgy.

COLOR PALETTE



#0A0A0C	#18181B	#B87333	#FFBF00	#D4AF37	#E2E8F0
Obsidian (Base)	Zinc (Surface)	Copper (Primary)	Amber (Active)	Gold (Success)	Slate (Text)

TYPOGRAPHY HIERARCHY

ROLE	FONT FAMILY	WEIGHT / SIZE	USAGE PATTERN
Data & Telemetry	`JetBrains Mono / Fira Code`	Regular / 12px - 14px	System logs, code outputs, status indicators, memory states.
Headings (H1-H3)	Segoe UI / Inter	Light (300) / 16px - 32px	Section titles, console headers, major status changes.
Body Copy	Segoe UI / Inter	Regular (400) / 14px	Prose, instructional text, capability descriptions.
Ornamental	Noto Sans Devanagari	Regular (400) / 10px - 24px	Subtle watermarks and bilingual subscripts (opacity < 30%).

ACCESSIBILITY & CONTRAST BOUNDARIES (GOOGLE M3)

Following Material 3 guidelines, pure drop-shadows are eliminated in favor of tonal borders and background contrasts. Interactive elements in the web console must maintain a minimum hit area of **48x48 dp**. Text contrast against the #0A0A0C background must meet the WCAG AA standard of 4.5:1. The Copper (#B87333) and Gold (#D4AF37) accents are specifically calibrated to pass contrast checks against the Zinc surface layers.